HDOC – 15th day

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Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

❖ PROBLEM STATEMENT:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
- Check whether all digits are prime.

○ Algorithm:

1) Start.

2) Declare variables num, choice, digit, i and an array freq[10].

3) Initialize all elements of freq[10] to 0.

4) Input a positive integer from the user.

5) Display the menu:

- 1 – Check whether all digits are unique.
- 2 – Find the frequency of each digit.

6) Input the user's choice.

7) Use switch(choice) to perform the selected operation.

7.1) If choice is 1:

- Extract each digit using num % 10.
- Increase the frequency of that digit.

- Remove the last digit using num / 10.
- Check the frequency of all digits.
- If any digit occurs more than once, display "Not Unique".
- Otherwise, display "Unique".

7.2) If choice is 2:

- Extract each digit using num % 10.
- Increase its frequency.
- Remove the last digit using num / 10.
- Display the frequency of digits from 0 to 9.

7.3) If the choice is neither 1 nor 2, display "Invalid Choice".

8) Stop.

- ## TEST CASE TABLE:

Test case	Input	choice	Expected output	Actual output	Result
1.	248	1	All digits are even	All digits are even	Pass
2.	867	1	All digits are not even	All digits are not even	Pass
3.	37	2	All digits are prime.	All digits are prime.	Pass
4.	21	2	All digits are not prime.	All digits are not prime.	Pass
5.	23	3	Invalid choice	Invalid choice	Pass

- ## C PROGRAM:

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Open

```c
#include <stdio.h>

int main()
{
    int num, n, digit, choice;
    int test = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\n1. Check whether all digits are even");
    printf("\n2. Check whether all digits are prime");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    n = num;

    switch(choice)
    {
        case 1:
            while(n > 0)
            {
                digit = n % 10;

                if(digit % 2 != 0)
                {
                    test = 0;
                    break;
                }

                n = n / 10;
            }

            if(test == 1)
                printf("All digits are even");
            else
                printf("All digits are not even");

            break;
```

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1 2 3 4

Open

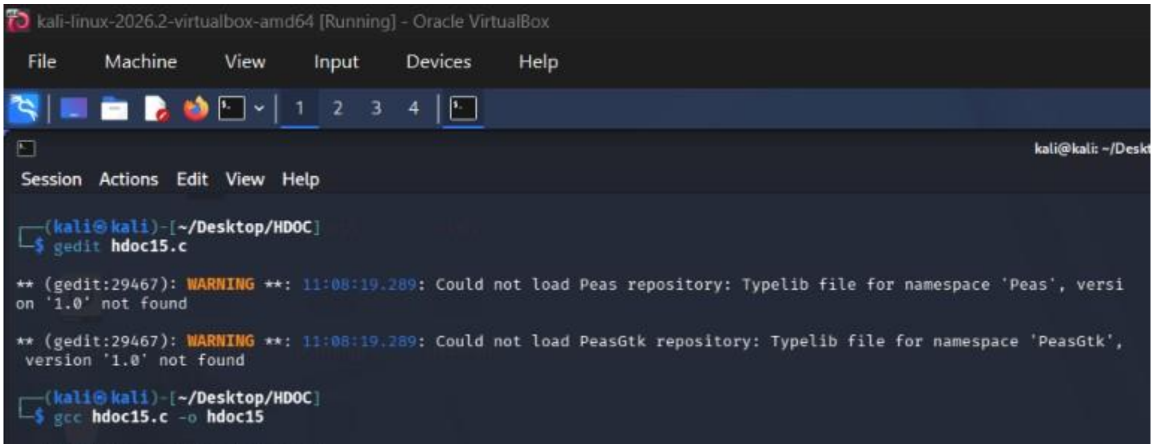
```
                n = n / 10;
            }

            if(test == 1)
                printf("All digits are even");
            else
                printf("All digits are not even");

            break;


        case 2:
            while(n > 0)
            {
                digit = n % 10;

                if(digit != 2 && digit != 3 &&
                   digit != 5 && digit != 7)
                {
                    test = 0;
                    break;
                }

                n = n / 10;
            }

            if(test == 1)
                printf("All digits are prime");
            else
                printf("All digits are not prime");

            break;


        default:
            printf("Invalid choice");
    }

    return 0;
}
```

Compile the code using:

gcc hdoc15.c -o hdoc15

Execute the code using:

./hdoc15



Here are some screenshots of the test cases of the code: